

Ankur Desai

(248) 657-3805 | ardusa05@gmail.com | www.ardusa.dev | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

May 2026 – Present

Undergraduate Research Assistant

East Lansing, MI

- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.
- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, then drove the decomposition attack to an 80% success rate against the per-step triage gate.

MindMosaic

Aug 2025 – May 2026

Software Engineering Intern

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Authored an AWS deployment script and managed the environment as code with Terraform, replacing a manual 30-minute multi-service bring-up with a single-command, version-controlled provision.

Claude Builder Club @ MSU

Oct 2025 – Present

Co-Founder & Vice President

East Lansing, MI

- Built event check-in as a single atomic database RPC that validates a QR token, records attendance, and awards points server-side, making door-time check-ins concurrency-safe and idempotent, sustaining 150+ check-ins per event.
- Shipped the platform continuously from main on a GitHub Actions pipeline that applies database migrations and deploys edge functions on every push, taking the team from manual changes to reviewed continuous deploys.

PROJECTS

Dadei | React, Python, Redis, Docker

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Root-caused the worker queue's silent exit-zero death on a Redis drop, then added a health check that detects it and a three-layer restart that recovers it, running the Python backend as two services from one Docker image.
- Routed all model-proposed side effects through one approval chokepoint with a cancelable countdown, giving every irreversible action one auditable seam between the LLM and the outside world.
- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Solved the same-room, multi-mic problem with two arbitration layers, a live utterance-owner election and a 400ms windowed quality election on SNR, clarity, and cadence, cutting duplicate transcription by 50%.

fliks | Go, Object Storage

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.

TECHNICAL SKILLS

Languages C++, Go, Java, Python, SQL, Bash

Frameworks Spring Boot, React

Databases PostgreSQL, Neo4j

Cloud & Infrastructure AWS (EC2, S3, RDS), Terraform, GitHub Actions